

Latika Liladhar Dekate

(669) 467-0836 | ldekate@asu.edu | LinkedIn | GitHub | Portfolio | CPT/OPT Eligible | **Available: Summer & Fall 2026**

EDUCATION

Arizona State University, Ira A. Fulton Schools of Engineering

M.S., Data Science, Analytics & Engineering (GPA: 3.89/4.00)

Tempe, AZ

Graduating Dec 2026

S.I.E.S. Graduate School of Technology

B.E., Electronics & Telecommunication Engineering (GPA: 3.71/4.00)

Mumbai, IN

Aug 2020 – Jun 2024

TECHNICAL SKILLS

AI & LLMs: LangChain, GraphRAG, Hugging Face Transformers (FinBERT, DistilBERT, Llama 3.1), RAG Pipelines, Agentic Workflows, Prompt Engineering

ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Feature Engineering, Anomaly Detection, Model Evaluation

Programming & Data: Python (NumPy, pandas, multiprocessing), SQL/PostgreSQL, PySpark, Bash, R

Data Engineering: ETL Pipelines, Apache Kafka, Neo4j, ChromaDB, Docker, GCP

Visualization & Tools: Streamlit, Flask, D3.js, Tableau, Power BI, Git, Linux, Kubernetes

PROJECTS

Stealthy Data Poisoning in RAG Systems | *Python, LangChain, ChromaDB, Llama 3.1*

Mar 2026 – Apr 2026

- Designed a controlled security evaluation framework over a 5,000-document RAG pipeline, testing 5 stealthy poisoning variants against 1,000 benchmark queries to quantify retrieval degradation under adversarial conditions.
- Measured near-random anomaly separability (AUC 0.527) across attack types, identifying Top-K capping as the most effective mitigation strategy and producing a reproducible evaluation protocol for future RAG security benchmarks.

AstraNode : GraphRAG Research Intelligence | *Neo4j, GraphRAG, LangChain, Python*

Oct 2025 – Nov 2025

- Architected a graph-based retrieval system over 600+ NASA research papers using Neo4j and LangChain, replacing keyword search with semantic graph and surfacing cross-mission connections previously inaccessible through traditional indexing.
- Reduced research discovery time by 40% across complex multi-domain queries; delivered full pipeline documentation for reproducibility and won the Arizona Regional NASA Space Apps Challenge.

Advanced Text Analyzer | *Flask, REST API, D3.js, Full-Stack*

Aug 2025 – Sep 2025

- Built a full-stack NLP application unifying 10,000+ records from CSV and JSON sources, deploying domain-specific transformer models (FinBERT for finance, DistilBERT for reviews) with 95%+ accuracy across three classification domains.
- Integrated Named Entity Recognition via SpaCy and a Flask REST API with a D3.js dashboard, providing stakeholders with traceable sentiment trends and severity metrics through a single interface.

EXPERIENCE

Arizona State University, DaRL Lab, SCAI

Research Intern, Traffic Signal Control (LibSignal)

Tempe, AZ

May 2026 – Present

- Sourcing and reviewing peer-reviewed RL literature to support the Related Work section of a faculty-led publication on the LibSignal benchmark, writing structured paper summaries for use in manuscript development.
- Synthesizing key contributions across methods including PressLight, CoLight, and MPLight, assessing their relevance and positioning relative to the benchmark framework under PhD student supervision.

Google Developer Student Clubs, S.I.E.S. Graduate School of Technology

ML & Data Science Program Coordinator

Mumbai, IN

Jul 2022 – Apr 2024

- Designed and delivered a 10-week ML curriculum covering PyTorch, TensorFlow, and production deployment practices for 150+ students, increasing project completion rates by 15% through structured mentoring and iterative feedback.
- Directed cross-functional student teams through the end-to-end ML lifecycle, from data preprocessing to model deployment, building a cohort capable of shipping production-grade projects independently.

S.I.E.S. Graduate School of Technology

Machine Learning & Image Processing Intern

Mumbai, IN

Jun 2022 – Jul 2022

- Diagnosed and debugged failure cases in an object detection pipeline by tuning normalization, cropping, and augmentation parameters in Python, improving held-out performance by 10% and reducing label noise in production review.
- Addressed class imbalance on a credit card fraud dataset by benchmarking sampling strategies and tree-based classifiers, selecting a final model achieving an F1 score of 0.92 aligned to product risk thresholds.

CERTIFICATIONS & RECOGNITION

Patent: Multilayer Microstrip Aerial Transmitter Receptor for Wi-Fi Applications

Apr 2024

NASA Space Apps Challenge : Regional Winner & Global Nominee

Oct 2025

Google Cloud Career Practitioners Certificate : Cloud Engineering & Data Science / ML Tracks

Jan 2023